ABHAY RAJESH PATLE

AI/ML & Cloud Engineer | Python • AWS • Machine Learning • Automation

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PROFESSIONAL SUMMARY

AI/ML & Cloud Engineer and B.Tech Computer Science student with hands-on experience building machine learning, Generative AI, backend, and AWS cloud solutions through academic, hackathon, and personal projects. Skilled in Python, PyTorch, XGBoost, FastAPI, Amazon Bedrock, AWS Lambda, IAM, and serverless architecture. Experienced in developing AI-powered applications, predictive models, cloud security automation, and REST APIs, with a strong foundation in machine learning pipelines, explainable AI, and MLOps practices.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Deep Learning, PyTorch, XGBoost, Scikit-Learn, Computer Vision, Explainable AI (SHAP/TreeSHAP), Feature Engineering, Predictive Modeling, Generative AI, RAG

Cloud & AWS: AWS Lambda, Amazon Bedrock, IAM, EC2, S3, VPC, RDS, DynamoDB, API Gateway, CloudWatch, Boto3, Serverless Architecture, Cloud Infrastructure, Cloud Security

Backend & APIs: Python, FastAPI, REST APIs, Asynchronous APIs, JWT Authentication, Rate Limiting, CORS, Input Validation, Node.js, Express.js

DevOps & Tools: Docker, Linux Systems Administration, Git, GitHub, CI/CD Pipelines, Automation

Databases & Data: PostgreSQL, MongoDB, SQL, ETL, Data Pipelines

Frontend & Analytics: React, Streamlit, Power BI, Data Analytics

Security: IAM Security, Least Privilege, Zero Trust, Security Auditing, Compliance Guardrails

PROJECTS

Nagpur Pulse — AI Traffic Risk & Dynamic Police Deployment Support — Lead AI/ML Engineer | *Hackathon Project — Viksit Nagpur Hackathon 2026, Team of 5*

Technologies: *Python, XGBoost, SHAP (Explainable AI), Scikit-Learn, FastAPI, Geospatial Analytics*

- Engineered the full AI/ML pipeline, transforming raw telemetry into structured risk factors from historical accident records, speed variance, traffic volume, and environmental conditions.
- Developed and hyperparameter-tuned a multivariate XGBoost model for predictive modeling of real-time accident risk scores across city road junctions.
- Implemented Explainable AI using TreeSHAP to deliver feature-level transparency and interpretable insights for law enforcement decision-making.
- Designed low-latency asynchronous FastAPI REST endpoints to serve real-time predictive scores and support dynamic police personnel dispatch recommendations.

Sahara — AI-Assisted Mental Healthcare & Wellness Platform — AI/ML & Full-Stack Engineer | *Final Year Project*

Technologies: *React, Node.js, Express.js, MongoDB, FastAPI, Generative AI, REST APIs, JWT, Pytest*

- Developed a full-stack AI-assisted wellness platform using React, Node.js, Express.js, and MongoDB, integrating mood tracking, PHQ-9/GAD-7 assessments, journaling, AI chat, and appointment management.
- Built a FastAPI-based Generative AI service with conversational context, risk-level classification, and automated crisis-risk detection backed by predefined safety responses and escalation workflows.
- Secured REST APIs with JWT authentication, refresh-token rotation, bcrypt password hashing, rate limiting, CORS, and input validation; validated AI workflows with Pytest.

IAM Guard AI Auditor v2.0 — Cloud Security & Serverless Developer | *Personal Project*

Technologies: *AWS Lambda, Amazon Bedrock, IAM, Python, Boto3, Serverless Architecture*

- Architected an event-driven serverless cloud security auditor that parses and evaluates IAM policy JSON documents in real time.

- Integrated Amazon Bedrock foundation models to calculate IAM security scores (0–100) and flag over-privileged or hazardous permissions.
- Automated least-privilege remediation templates with side-by-side policy comparisons to streamline cloud security review workflows.
- Enforced automated cloud compliance guardrails and Zero-Trust identity controls using Boto3 and AWS Lambda.

AI Plant Identifier & Health Analytics — ML Developer | *Personal Project*
Technologies: *Python, PyTorch, ResNet18, Computer Vision, Streamlit Cloud*

- Built an end-to-end deep learning computer vision pipeline using a custom-trained ResNet18 model for plant species and disease identification.
- Implemented Top-5 confidence score analytics and detailed botanical taxonomy classification.
- Developed automated diagnostic PDF report generation and deployed the interactive application to Streamlit Cloud.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering
Tulsiramji Gaikwad-Patil College of Engineering & Technology (TGPCET), Nagpur | SGPA: 8.3 | 2023–2027

Higher Secondary Certificate (HSC) — Science
Late Khushal Kapse Junior College, Gondia | 71.50% | 2022–2023

Secondary School Certificate (SSC)
Adasi Tanda High School, Adasi | 77.00% | 2020–2021

CERTIFICATIONS & VIRTUAL INTERNSHIPS

- AWS Academy: Cloud Foundations, Cloud Architecting
- EduSkills Virtual Internships: AWS AI-Powered Cloud Engineer; AWS Data Engineering; Google AI-ML Virtual Internship
- FutureSkills Prime: Fundamentals of Amazon Braket (Quantum Computing); Handling Sensitive Data on the Cloud; Applications of Big Data Analytics (BDA); Popular Software Tools Used in Cloud Computing
- Microsoft, Nirmaan Org & Redington Foundation: 180-Hour AI/ML Intensive Training Program & Hands-on Capstone

COMMUNITY CONTRIBUTIONS & ACHIEVEMENTS

- Viksit Nagpur Hackathon 2026: Finalist/Participant with project Nagpur Pulse (organized by Jan Manthan Foundation, RTMNU & VNIT Nagpur).
- AWS Student Community Day Nagpur 2026 (IIM Nagpur): Official Certificate of Participation; built IAM security solutions inspired by community sessions.
- Google I/O Extended Nagpur 2026 (Infosys MIHAN): Participated in technical tracks on Agentic SRE, Gemini production deployment, and developer challenges.
- AWS Educate Emerging Talent Community (ETC): Active contributing member.